

Mathematics for Computing

Assignment 1

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Sets $\longrightarrow$ Relations $\longrightarrow$ Structure

Cardinality of a Set

Cardinality is the number of distinct elements contained in a set.

$$|A| = \text{number of elements of } A$$

Example

A computing-oriented example

Let

$$A = \{\text{CPU, GPU, RAM, SSD, NIC, USB}\}.$$

There are six distinct components, so

$$|A| = 6.$$

Types of Sets: Finite Set

A **finite set** contains a limited number of elements.

$$|A| = n, \quad n \in \mathbb{N}.$$

Example

Example: network status labels

A monitoring system may use

$$S = \{\text{Online}, \text{Idle}, \text{Busy}, \text{Offline}\}.$$

The number of possible labels is four, hence S is finite.

$$S = \{x \in U \mid x \text{ is one of the four status labels}\}.$$

Types of Sets: Infinite Set

An **infinite set** has infinitely many possible elements.

Example

Example: possible latency values

If latency is represented by a non-negative real number,

$$L = \{x \in \mathbb{R} \mid x \geq 0\}.$$

Between any two different real-valued measurements, more real values can be chosen. Thus L is infinite.

$$L = \{x \in \mathbb{R} \mid x \geq 0\}.$$

Types of Sets: Null Set

A **null (empty) set** contains no elements and is denoted by $\emptyset$.

Example

Example: unavailable error category

Suppose a test run records only errors from the set

$$E = \{\text{timeout, syntax, memory}\}.$$

If a particular run contains no memory errors, then

$$M = \{\text{memory errors observed in the run}\} = \emptyset.$$

Types of Sets: Countable Set

A **countable set** can be listed as a sequence and associated with natural numbers.

Example

Example: packet sequence numbers

For a finite capture, packet identifiers can be listed:

$$P = \{101, 102, 103, 104, 105, \dots\}.$$

The identifiers can be indexed one after another, so the set is countable.

$$P = \{f(n) \mid n \in \mathbb{N}\}$$

for an appropriate enumeration function f .

Types of Sets: Power Set

The **power set** $\mathcal{P}(A)$ is the set of every subset of A , including $\emptyset$ and A itself.

Example

Example: system capabilities

Let

$$C = \{\text{GPS}, \text{WiFi}, \text{Bluetooth}\}.$$

Then

$$\mathcal{P}(C) = \{\emptyset, \{\text{GPS}\}, \{\text{WiFi}\}, \{\text{Bluetooth}\}, \\ \{\text{GPS}, \text{WiFi}\}, \{\text{GPS}, \text{Bluetooth}\}, \{\text{WiFi}, \text{Bluetooth}\}, C\}.$$

$$|\mathcal{P}(C)| = 2^{|C|} = 2^3 = 8.$$

Types of Sets: Subset

A set T is a **subset** of D if every element of T also belongs to D .

$$T \subseteq D.$$

Example

Example: selected software tools

$$D = \{\text{Python, Java, C, JavaScript}\}, \quad T = \{\text{Python, JavaScript}\}.$$

Every element of T occurs in D , therefore

$$T \subseteq D.$$

Types of Sets: Equal Set

Two sets are **equal** when they contain exactly the same elements, regardless of order.

Example

Example

$$A = \{\text{TCP}, \text{UDP}, \text{HTTP}\}$$

and

$$B = \{\text{HTTP}, \text{TCP}, \text{UDP}\}.$$

Since order does not matter for sets,

$$\boxed{A = B}.$$

Types of Sets: Equivalent Set

Two sets are **equivalent** if they contain the same number of elements.

Example

Example

Let

$$A = \{\text{Linux, Windows, macOS}\}$$

and

$$B = \{\text{Java, Python, C++}\}.$$

Both have three elements:

$$|A| = |B| = 3, \quad \therefore \quad \boxed{A \sim B}.$$

Types of Sets: Superset

A set D is a **superset** of T when it contains every element of T .

$$D \supseteq T.$$

Example

Example

$$D = \{\text{HTML}, \text{CSS}, \text{JavaScript}, \text{React}\}$$

and

$$T = \{\text{HTML}, \text{CSS}\}.$$

Hence

$$\boxed{D \supseteq T}.$$

Types of Sets: Singleton Set

A **singleton set** contains exactly one element.

$$|A| = 1.$$

Example

Example: one active server

If only one server is currently selected for deployment,

$$A = \{\text{server-07}\}.$$

Therefore A is a singleton set.

$$A = \{x \mid x = a\}.$$

Set Operations

Set operations are used to combine or compare sets.

1. Union

The union of A and B contains all elements belonging to A or B .

$$A \cup B = \{x \mid x \in A \text{ or } x \in B\}$$

2. Intersection

The intersection contains elements common to both sets.

$$A \cap B = \{x \mid x \in A \text{ and } x \in B\}$$

3. Difference

The difference $A - B$ contains elements in A but not in B .

4. Complement

The complement of A contains elements in the universal set U that are not in A .

$$A^c = U - A$$

5. Symmetric Difference

The symmetric difference contains elements belonging to exactly one of the two sets.

$$A \triangle B = (A - B) \cup (B - A)$$

Important

Set operations are useful for organizing, filtering, and comparing data in computing and data science.

Set Operations: Computing Example

Consider two sets of programming languages:

$$A = \{\text{Python, Java, C}\}, \quad B = \{\text{Java, C++, Python}\}$$

Union

$$A \cup B = \{\text{Python, Java, C, C++}\}$$

Intersection

$$A \cap B = \{\text{Python, Java}\}$$

Difference

$$A - B = \{\text{C}\}$$

Set-builder Form: Examples I

1. Natural numbers that are multiples of 4 but not 8

$$A = \{x \in \mathbb{N} \mid 4 \mid x, 8 \nmid x\}.$$

$$A = \{4, 12, 20, 28, 36, \dots\}.$$

2. Perfect fourth powers

$$B = \{x \in \mathbb{N} \mid x = n^4, n \in \mathbb{N}\}.$$

$$B = \{1, 16, 81, 256, 625, \dots\}.$$

Set-builder Form: Examples II

3. Natural numbers between 12 and 60

$$A = \{x \in \mathbb{N} \mid 12 < x < 60\}.$$

$$A = \{13, 14, 15, \dots, 59\}.$$

4. Positive integers leaving remainder 2 when divided by 7

$$B = \{x \in \mathbb{Z}^+ \mid x = 7n + 2, n \in \mathbb{N}_0\}.$$

$$B = \{2, 9, 16, 23, 30, \dots\}.$$

Set-builder Form: Examples III

5. Even natural numbers from 2 to 20

$$A = \{x \in \mathbb{N} \mid x = 2n, 1 \leq n \leq 10\}.$$

$$A = \{2, 4, 6, 8, 10, 12, 14, 16, 18, 20\}.$$

6. Factors of 36

$$B = \{x \in \mathbb{N} \mid x \mid 36\}.$$

$$B = \{1, 2, 3, 4, 6, 9, 12, 18, 36\}.$$

Set-builder Form: Examples IV

7. Ordered pairs whose product is 12

$$A = \{(x, y) \in \mathbb{Z}^2 \mid xy = 12\}.$$

Some positive pairs are

$$(1, 12), (2, 6), (3, 4), (4, 3), (6, 2), (12, 1).$$

8. Natural numbers divisible by 4 or 5

$$B = \{x \in \mathbb{N} \mid 4 \mid x \text{ or } 5 \mid x\}.$$

$$B = \{4, 5, 8, 10, 12, 15, 16, 20, \dots\}.$$

Set-builder Form: Examples V

9. Cubes of natural numbers

$$A = \{n^3 \mid n \in \mathbb{N}\}.$$

$$A = \{1, 8, 27, 64, 125, \dots\}.$$

10. Prime numbers from 1 to 35

$$B = \{x \in \mathbb{N} \mid x \text{ is prime and } 1 < x \leq 35\}.$$

$$B = \{2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31\}.$$

Inclusion–Exclusion Principle

The **Inclusion–Exclusion Principle** counts elements in a union while correcting for overlaps.
For three sets:

$$|A \cup B \cup C| = |A| + |B| + |C| - |A \cap B| - |A \cap C| - |B \cap C| + |A \cap B \cap C|$$

Example

Small numerical example

Suppose $|A| = 24$, $|B| = 18$, and $|A \cap B| = 7$. Then

$$|A \cup B| = 24 + 18 - 7 = \boxed{35}.$$

The intersection is subtracted because those elements were counted twice.

Partially Ordered Sets (Posets)

A **partially ordered set** is a set together with a relation satisfying:

- 1 **Reflexive:** $a \preceq a$.
- 2 **Antisymmetric:** if $a \preceq b$ and $b \preceq a$, then $a = b$.
- 3 **Transitive:** if $a \preceq b$ and $b \preceq c$, then $a \preceq c$.

Example

Example: divisibility

Let

$$P = \{1, 2, 4, 8\}$$

with the relation “divides” ($|$). Since $1 \mid 2$, $2 \mid 4$, and $4 \mid 8$, the relation forms a poset.

Hasse Diagram: Divisibility Poset

Consider

$$P = \{1, 2, 4, 8\}, \quad \text{relation } |.$$

Only the immediate covering relations are drawn.



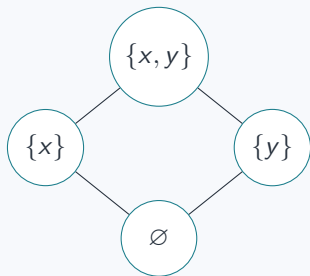
Thus $(P, |)$ is a partially ordered set.

Hasse Diagram: Subset Poset

Let

$$P = \{\emptyset, \{x\}, \{y\}, \{x, y\}\}$$

with relation $\subseteq$.



The diagram represents the subset relation without drawing transitive edges.

Hasse Diagram: Factor Poset

Consider

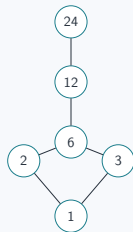
$$P = \{1, 2, 3, 6, 12, 24\}$$

under divisibility.

The immediate covering relations are

$$1 \mid 2, \quad 1 \mid 3, \quad 2 \mid 6, \quad 3 \mid 6, \quad 6 \mid 12, \quad 12 \mid 24.$$

These relations are enough to construct the Hasse diagram.



Chain and Antichain

Chain

Definition: Every pair of elements is comparable.

$$a \preceq b \quad \text{or} \quad b \preceq a$$

Example:

$$P = \{1, 2, 4, 8\}, \quad C = \{1, 2, 4\}$$

Under divisibility,

$$1 \mid 2, \quad 2 \mid 4, \quad 1 \mid 4.$$

Hence,

C is a chain

Antichain

Definition: No two distinct elements are comparable.

$$a \not\preceq b \quad \text{and} \quad b \not\preceq a$$

Example:

$$P = \{1, 2, 3, 6\}, \quad A = \{2, 3\}$$

Under divisibility,

$$2 \nmid 3, \quad 3 \nmid 2.$$

Hence,

A is an antichain